JNTUH Used Paper 28-11-2020A

Code No: 124AG **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) Design a Moore Machine to determine the residue mod 4 for each binary string treated as integer.
b) Construct a deterministic finite automaton accepting all strings over {0, 1} having odd number of 0's and 1's. [8+7]

2. Minimize the following DFA shown in figure 1. [15]

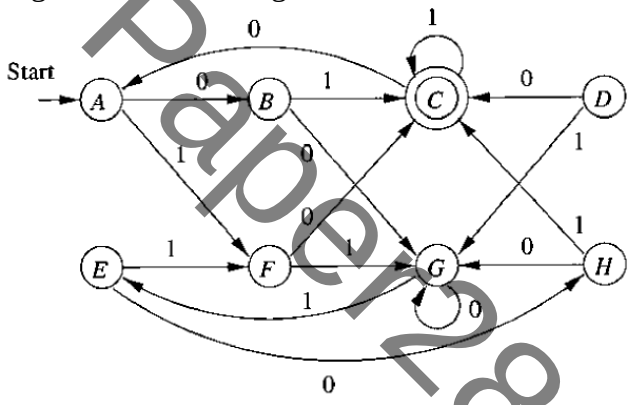


Figure: 1

3. Write the algorithm for converting FA to Regular Expressions. [15]

4. Convert the given FA to Regular Expression using Arden's theorem shown in figure 2. [15]

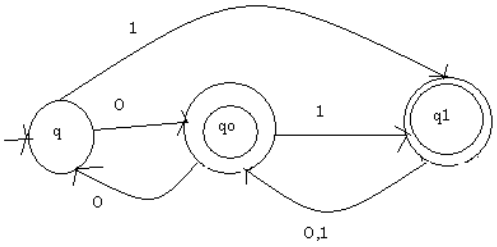


Figure: 2

5.a) Design PDA for the language L = $\{a^m b^m c^m \mid m \geq 1\}$.
b) Convert the following CFG to PDA: S → aSbb | aab [7+8]

6.a) Construct the PDA to the following grammar:
S→AB
A→BS/b
B→SA/a

b) Convert the following grammar in to GNF: [7+8]
S→ABA/AB/BA/AA/B
A→aA/a
B→bB/B

7. Design a T.M for copying of information from one place to the other place. Make all the necessary assumptions. Give Example of the working of your T.M. [15]

8. Describe in detail about Turing reducibility and Halting Problem. [15]

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